VALUE ADDED PROBLEM 1:

Given a Binary Search Tree and a range [low, high]. Find all the numbers in the BST that lie in the given range. Note: Element greater than or equal to root go to the right side. Concept: Binary Search Tree

Example 1:

Input:

```
        17
      /    \
     4      18
   /   \
  2     9
```

l = 4, h = 24

Output:

4 9 17 18

Example 2:

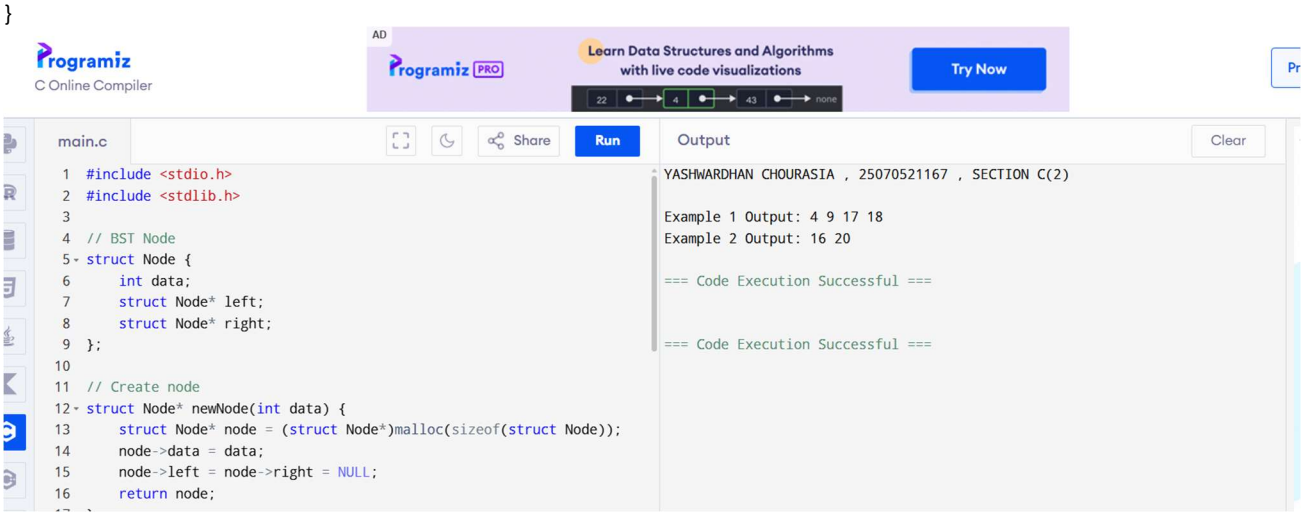
Input:

```
        16
      /    \
     7      20
   /   \
  1     10
```

l = 13, h = 23

Output:

16 20

```
#include <stdio.h>
#include <stdlib.h>

// BST Node
struct Node {
   int data;
   struct Node* left;
   struct Node* right;
};

// Create node
struct Node* newNode(int data) {
   struct Node* node = (struct Node*)malloc(sizeof(struct Node));
   node->data = data;
   node->left = node->right = NULL;
   return node;
}

// Insert into BST
struct Node* insert(struct Node* root, int data) {
   if (root == NULL)
      return newNode(data);

   if (data < root->data)
      root->left = insert(root->left, data);
   else
      root->right = insert(root->right, data);

   return root;
}

// Print values in range
void printInRange(struct Node* root, int low, int high) {
   if (root == NULL) return;
```

```
    if (root->data > low)
        printInRange(root->left, low, high);

    if (root->data >= low && root->data <= high)
        printf("%d ", root->data);

    if (root->data < high)
        printInRange(root->right, low, high);
}

int main() {

    printf("YASHWARDHAN CHOURASIA , 25070521167 , SECTION C(2)\n\n");

    // -------- Example 1 --------
    struct Node* root1 = NULL;
    int arr1[] = {17, 4, 18, 2, 9};
    int n1 = 5;

    for (int i = 0; i < n1; i++)
        root1 = insert(root1, arr1[i]);

    printf("Example 1 Output: ");
    printInRange(root1, 4, 24);
    printf("\n");

    // -------- Example 2 --------
    struct Node* root2 = NULL;
    int arr2[] = {16, 7, 20, 1, 10};
    int n2 = 5;

    for (int i = 0; i < n2; i++)
        root2 = insert(root2, arr2[i]);

    printf("Example 2 Output: ");
    printInRange(root2, 13, 23);
    printf("\n");

    printf("\n=== Code Execution Successful ===\n");

    return 0;
}
```